

Response Letter for Manuscript MKSC-2025-0013

Dear Senior Editor Tucker, Associate Editor, and Reviewers,

We are grateful for the opportunity to revise this manuscript and for the editors' and reviewers' careful and constructive engagement throughout this process. We have conducted a thorough revision that substantially strengthens both the empirical identification and the contribution of the paper.

Note to Referees on the Revision Process. This revision followed an extended editorial dialogue. Following the first-round reports, the Senior Editor suggested that we first reframe the contribution before undertaking the full empirical revision, to ensure the paper's direction was sound before extensive additional work was done. We exchanged three rounds of proposals and feedback with the editors on the revised framing and scope of the revision. This process resulted in a final revision contract agreed with the Associate Editor and Senior Editor, which established the following structure for the revision:

- (1) A demand-side main analysis with strengthened identification (addressing the major empirical concerns raised by R1, R2, and the AE);
- (2) A new demand-side heterogeneity analysis using transcript-derived content features to examine which content characteristics are associated with larger or smaller engagement graph effects; and
- (3) A supply-side difference-in-differences analysis examining whether creators alter their content supply decisions (along margins that do not require video transcripts) following the platform-wide rollout of engagement graphs.

We flag this process upfront because referees may notice that the paper’s structure and emphasis have changed substantially relative to the first-round submission. These changes reflect both the editors’ guidance and our own assessment. We believe the revised paper is substantially stronger on all dimensions and we hope referees agree.

Summary of New Empirical Analyses. The revision involves a large number of new analyses. To help navigate the response, Table 1 provides an overview of all new analyses, the referee comments they address, and a brief summary of findings.

Table 1: Summary of New Analyses in the Revision

Analysis	Addresses	Key Finding
<i>(1) Demand-side: Strengthening Identification</i>		
Upload weekday/weekend exact match	R1.1a, R2.3b	Effects are robust
Adjacent upload date matching	R1.1a, R2.3b	Effects are robust
Unmatched sample	R1.1c	Effects are robust
Fuzzy RD at 50K-view threshold	R1.1b	Effects are positive but insignificant (underpowered; completely different estimand)
Extended pre-treatment window	AE 1.3	More pre-trend periods; all near zero. Post-treatment effects robust
Placebo test (artificial timing)	AE 1.3	All placebo estimates are insignificant
<i>(2) Demand-side: Heterogeneous Treatment Effects</i>		
Segment-level content feature validation	AE Contribution	Content feature distributions match intuitive patterns; substantive features associated with higher engagement
Transcript-based content features as continuous moderators	AE Contribution, R1.2, R2.2b, R2.4b	Informational features (payoff lateness, info density, percentage of new concepts) moderate treatment effect; structural feature (narrative completeness) does not
<i>(3) Supply-side: Creator Content Decisions</i>		
DiD on upload volume & duration	AE Contribution	No significant effects on posting frequency or video length
DiD on content structure	AE Contribution	Largely null; marginal decline in conclusion segment share ($p = 0.054$)

We organize the response as follows, in order: Reviewer 1, Reviewer 2, and the Associate Editor. Each point from each reviewer and the AE is reproduced, followed immediately by our response. Where multiple reviewers raise the same concern, we note this and refer

forward or backward to avoid repetition.

Reviewer 1

Identification

R1, Identification: *A limitation of the identification strategy is that it does not fully control for potential time trends in video posting date. By design, treated and control videos are uploaded on different dates. Within each batch, earlier and later posted videos may not be comparable if underlying market condition differs by posting date, and if the shocks in earlier periods persist or are amplified over time by Youtube’s algorithm. If video posting dates are random, this would not be a concern, but certain creators may strategically choose upload dates (e.g., weekends vs. weekdays), leading to potentially systematic differences in video quality between treatment and control groups, conditional on having similar growth trajectories. Across batches, control videos are disproportionately from March due to the long time spans shown in Figure 6. If videos posted in different months have different time trends, or a different set of creators are posting (e.g. students posting more during spring breaks), then the estimate could be biased.*

Response: We thank Reviewer 1 for raising this concern carefully. We address both dimensions (upload weekday/weekend differences and macro-temporal confounding) with dedicated robustness checks, described below under R1.1a and R1.1b respectively.

R1.1a: *Alternative samples: the authors could run analyses using alternative samples that match on upload dates in different ways. For example, they could match videos based on adjacent uploading dates, and/or based on only weekday vs. weekends.*

Response: We implement both approaches.

Weekday/weekend matching. A potential threat to our identification strategy arises from systematic differences in upload day between treated and control groups. By design,

treated and control videos are uploaded on different dates within and across batches. If certain creators systematically prefer uploading on weekends versus weekdays, and if weekend versus weekday uploads follow systematically different engagement trajectories over the video lifecycle in a way that cannot be detected by our pre-trend test, then an imbalance in upload day composition between treated and control groups could violate the parallel trends assumption underlying our DiD strategy.

To address this concern directly, we augment our baseline CEM procedure with an exact-match requirement on upload day (weekday vs. weekend), ensuring that treated and control videos within each matched stratum share the same upload-day pattern while keeping all other matching variables identical. The resulting sample contains 1,843 treated and 1,390 control videos. Both CS and TWFE estimates closely mirror our baseline results, indicating that upload day timing does not meaningfully confound our main estimates. These results are reported in Tables [X] and [X] (Robustness to Upload Day Timing) in the revised paper.

Adjacent upload date matching. Another threat to our identification arises from macro-temporal confounding. By design, treated videos — those with shorter time-till-threshold — tend to be uploaded in February and April, while control videos with longer time-till-threshold are disproportionately from March uploads. While our video fixed effects absorb time-invariant video-level differences, time-varying macro-level shocks that differentially affect videos uploaded in systematically different calendar periods could in principle threaten the parallel trends assumption. Our event study analyses in Section [X] already provide supporting evidence against this concern, showing no significant pre-treatment differential trends between treated and control videos. Nonetheless, to more directly rule out macro-temporal confounding, we implement an additional robustness check that restricts comparisons to videos uploaded close together in calendar time.

Specifically, we impose a temporal proximity filter on top of our baseline CEM procedure. After CEM matching, we retain within each matched stratum only those videos that have at least one counterpart from the opposite treatment group uploaded within seven calendar

days. This ensures that all treated-control comparisons in our estimation sample are made between videos uploaded close together in calendar time, making it implausible that macro-level temporal trends are driving the estimated treatment effects.

This filter substantially reduces the sample size by construction, retaining only videos uploaded near the batch boundaries between treated and control calendar periods — specifically, treated videos uploaded at the end of February and beginning of April, and control videos uploaded at the beginning and end of March. The temporally filtered sample contains 477 treated and 475 control videos, approximately 22% of the baseline matched sample.

Despite this reduction in statistical power, the results remain directionally consistent with our main findings across both estimators and all three outcomes. These results are reported in Tables [X] and [X] (Robustness to Macro-Temporal Confounding) in the revised paper.

Together, these checks provide strong evidence that upload-day composition and cross-month temporal trends do not drive our main findings.

***R1.1b:** Alternative RD design: the authors could explore the other RD based on the viewership cutoff. The sample would contain videos uploaded on adjacent dates (or week-day/weekends), and are within a narrow window above and below the viewing threshold by the batch processing date. The assumption here is that videos around the 50K viewer threshold are similar in quality.*

Response: As an alternative identification strategy, we exploit the 50,000-view eligibility threshold using a regression discontinuity (RD) design. This approach is conceptually distinct from our main DiD strategy: rather than leveraging quasi-random variation in batch processing timing, it compares videos that barely cross the view threshold against otherwise similar videos just below it. Importantly, the two strategies estimate different causal quantities: the DiD recovers the ATT for the broader population of videos receiving engagement graphs during their first two weeks, while the RD recovers the Local Average Treatment

Effect (LATE) for compliers at the 50,000-view margin. We therefore present this as an additional lens on the data rather than a direct replication of the main results.

We implement a fuzzy RD design using cumulative views measured one period prior to the eligibility date as the running variable, centered at 50,000. We use views one period prior rather than at the eligibility period itself to avoid endogeneity, since views measured at the eligibility period could be mechanically inflated by the treatment. We restrict the sample to videos within a $\pm 2,000$ -view bandwidth around the cutoff and aged 2–7 days at the pre-eligibility period, yielding 467 videos below and 481 above the cutoff. The outcome variables are log new views, log new likes, and new comments aggregated over the two 48-hour periods following the eligibility date, consistent with the period definition and log transformation used in our main analysis. We estimate the LATE using `rdrobust` (Calonico et al., 2014) in STATA with its default optimal bandwidth selector, controlling for video age and view growth rate. We verify that there is no evidence of manipulation at the cutoff using the density test of Cattaneo et al. (2020), which fails to reject the null of no discontinuity in the running variable density, consistent with creators not strategically sorting their view counts around the threshold.

The fuzzy RD yields positive point estimates across all three outcomes but statistically insignificant results, which we attribute primarily to limited statistical power (approximately 150 effective observations in the optimal bandwidth on each side of the cutoff). Importantly, the RD and DiD strategies estimate conceptually distinct quantities: the RD recovers the LATE for compliers at the 50,000-view boundary, while the DiD recovers the ATT for the broader treated population. We therefore present this as a complementary identification lens rather than a direct replication, and report results in Table [X] and Figure [X] of the revised paper.

***R1.1c:** Robustness using unmatched samples: The event study graphs are helpful and increase our confidence in the identification strategy. The authors could also present*

results using the full, un-matched sample to show how matching affects the estimates. This would give us a better sense of how robust the main results are.

Response: We re-estimate our main specifications on the full, unmatched sample (10,058 videos) without applying any CEM procedure. Results closely mirror our baseline: both CS and TWFE estimates are positive and statistically significant. These results confirm that the main findings are not an artifact of the matching procedure and are reported in Tables [X] and [X] of the revised paper.

Mechanism

R1, Mechanism: *While observational learning is a plausible mechanism, I'm not sure that the heterogeneity analysis in Section 6 based on the number of peaks provides direct evidence for it.*

R1.2a. *The relationship between the number of peaks and increased views (Table 10, Column 1) may be partially mechanical. Some users may simply want to go through all the peaks in a video (not necessarily because they are learning from them), and this process takes longer when there are more peaks.*

R1.2b. *Some users may believe that the number of peaks could be informative about video quality – videos that have more peaks are “denser” and may contain more valuable information, which naturally attracts more viewing and likes. In this case, the heterogeneity may reflect differences in video quality rather than learning from graphs.*

R1.2c. *Similar argument applies to the result on increasing concentrated engagement distribution – engagement peaks likely correlated with higher quality for that video segment, which may cause users to replay those segments. If so, the increasing concentration may not entirely reflect users learning but may also reflect inherent segment quality.*

Response: Reviewer 1 raises important and well-taken points about the identification

challenges in the original observational learning and peak-based heterogeneity analysis. We agree with these critiques. In the revised paper, we have entirely removed the original Section 6 mechanism analysis, including the heterogeneity by number of peaks and the engagement concentration results. This decision reflects both the reviewers' concerns and the editorial guidance we received during the pre-revision dialogue, where the editors encouraged us to pivot toward a more ambitious and interpretable heterogeneity analysis based on video transcripts.

In the new heterogeneous treatment effects analysis, we extract informational features (payoff lateness, information density, percentage of new concepts) and a structural feature (narrative completeness) from video transcripts, and include all four as continuous moderators of the treatment effect. The key finding is that informational features significantly moderate the engagement graph's effect in predicted directions, while narrative structure does not. We describe this analysis in detail in the response to the AE's Contribution section below (p. [X]).

Reviewer 2

1. What drives causal effects

***R2.1a.** Per my reading, the paper finds some evidence for the causal effects of the engagement graph; e.g. in Figure 8 there is a jump in the number of views and likes of videos once they get an engagement graph. There are weak pre-trends but also a clear effect in subfigures (a) and (b), and a more gradual change in subfigure (c).*

Response: We appreciate Reviewer 2's reading. The revised paper strengthens the pre-trend evidence with an extended pre-treatment window analysis and a placebo test with artificially assigned treatment timing, which together more firmly establish that the post-

treatment jump is not an artifact of pre-existing differential trends. See our response to AE 1.3 for details.

R2.1b. *However, I do not think that there is convincing evidence that this effect is only due to the direct effect of the engagement graph feature and is not reinforced by YouTube’s algorithm. The paper argues that effects are not driven by algorithmic recommendations of the videos because (a) engagement graph appears one week after the video was uploaded and after it got the majority of views, and (b) the treatment effect size was small (the end of page 37). I disagree with this argument. First, changes in the number of views, likes, and comments, speed of these changes, and potentially the addition of the engagement feature itself can be part of the recommendation algorithm (which is not disclosed by YouTube). Second, given that the estimated effect size is small, even small adjustments in the algorithm may be driving it.*

Response: We agree with Reviewer 2 that our treatment effect plausibly captures both direct viewer responses to the engagement graph and indirect effects mediated by YouTube’s recommendation algorithm, and we explicitly acknowledge this in the revised paper. Importantly, we argue that the combined effect, i.e., direct viewer response plus any algorithmically mediated downstream effect, is the managerially relevant estimand. From the perspective of platforms and content creators, what matters is the total effect of the engagement graph’s activation on video performance, taking as given the current market conditions and algorithms. We frame the combined effect as such in the revised paper. We acknowledge that the exact magnitude of the estimated effects may be specific to YouTube’s recommendation algorithm at the time of our study, and that different algorithmic regimes could yield different magnitudes of indirect effects. We discuss this limitation in generalizability explicitly in the revised discussion section.

R2.1c. *I find it hard to interpret the estimates as a direct effect of the engagement feature on viewers because viewers learn that engagement feature is available only after they land on the video. If viewers get more utility from the product due to the engagement feature, I would expect that the overall viewership of videos should not change, but the number of likes and comments goes up. Instead, the paper finds the effect on both the number of viewers and the number of likes that are similar in magnitude.*

Response: This is an important interpretive point also raised by the AE (AE 1.1). The key lies in the definition of a “view” on YouTube: a view is counted when a user cumulatively watches at least 30 seconds of the video, including cases where a viewer skips ahead but accumulates 30 seconds of total watch time. The engagement graph, which is visible to a viewer once they have opened the video, may encourage viewers who would otherwise abandon the video early to stay longer or navigate to engaging segments, thereby crossing the threshold and generating an incremental view count.

Following AE’s suggestion (AE 1.1), we now highlight this retention interpretation explicitly earlier in the paper. We clarify that our view count results capture the effect on viewers crossing the 30-second threshold, which is itself a meaningful measure of consumption and viewer retention.

2. Interpretation of the estimates for video producers and YouTube

R2.2a. *The paper interprets the estimates as incremental engagement for YouTube, arguing that because the effect size is small and most viewership on YouTube happens on videos without engagement graphs (the middle of page 37), there are “minimal displacement effects” of the time people spend on other videos on YouTube. I do not understand this argument; viewers can reallocate their time to a video from other videos on YouTube or from the outside (non-YouTube) option, and even if view videos have engagement*

graphs they can (and likely do) crowd out viewership from other videos. There is no reason to assume that videos on YouTube do not compete with each other for the viewers' attention, which the current interpretation of the estimates does. If I had to state my prior on the nature of the effect that the paper finds, it would be the opposite: that most of the incremental effects that the paper estimates is driven by viewers reallocation of time within YouTube (also because the engagement graph and small changes in engagement are picked up by the recommendation algorithm).

Response: We appreciate Reviewer 2 pushing us on this point. We want to clarify that our original argument about “minimal displacement effects” was part of our SUTVA discussion: it was intended to argue that the treatment-induced increase in views on treated videos is too small in aggregate to meaningfully shift views *away from the specific control videos in our estimation sample*, which is what SUTVA requires. We were not making a broader claim that engagement graphs generate net new engagement for the platform as a whole. That said, we agree that the original language did not draw this distinction sharply enough, and we have revised accordingly. In the revised paper, we remove the claim of “minimal displacement effects” from the results discussion and instead confine the argument to its original SUTVA context: that the small magnitude of treatment-induced viewing changes relative to overall platform activity makes it unlikely that our control videos’ outcomes are materially affected by the treatment, which is the specific assumption our identification strategy requires. We do not make claims about platform-level welfare or whether the engagement graph generates incremental versus redistributed engagement at the platform level, as resolving this question would require platform-level data on total user time allocation that is beyond the scope of our study.

R2.2b. *The effects are sizable and important for video producers, but they do not have any control over the engagement graph feature (beyond trying to reach 50K views as soon as possible and potentially game the “batch processing” system). As a result, managerial*

implications are not clear.

Response: We agree that creators have limited direct control over the engagement graph itself. The revised paper addresses this concern through two new analyses that yield actionable implications despite this constraint.

First, our transcript-based heterogeneous treatment effects analysis shows that the engagement graph’s demand-side effect is systematically moderated by content characteristics that are within the creator’s control at the production stage. Specifically, we find that the engagement graph matters more when content is informationally lighter, when value delivery is deferred later in the video, and when new material is introduced more frequently throughout—each a condition under which a prospective viewer faces greater uncertainty about whether the remaining content is worth watching. While creators cannot control whether the engagement graph appears, these findings identify content dimensions along which the feature’s impact varies, and these dimensions reflect production choices that creators actively make (how to pace information delivery, where to place key payoffs, how to distribute novel material across the video). We are careful to note that these are *associations* between content characteristics and treatment effect magnitudes rather than causal prescriptions for content design, but they provide empirically grounded directions that creators can evaluate in their own production decisions. See Section [X] of the revised paper for full results.

Second, the supply-side analysis directly investigates whether creators adjust their production behavior in response to engagement graph availability. The largely null finding is itself informative for platform managers: it suggests that changes to the viewer-facing information environment alone may not be sufficient to shift creator behavior, and that direct incentive mechanisms (such as revenue sharing arrangements) may be more effective levers for guiding content production.

***R2.2c.** Separately, the engagement graph is more useful for people who want to skip ahead and watch only portions of the video. Are these viewers equally important for content producers compared to regular viewers? E.g. in terms of expected advertising revenue and donations.*

Response: This is a thoughtful question. We note that the relevant comparison for interpreting our estimates is not between “skipping” viewers and “regular” viewers, but between retaining a marginal viewer and losing that viewer entirely. As discussed in our response to R2.1c and AE 1.1, the engagement graph’s effect on view counts operates through a retention mechanism: viewers who would otherwise abandon the video before accumulating 30 seconds of watch time are encouraged to stay. In this sense, the incremental viewers our estimates capture are unambiguously more valuable than the counterfactual of no engagement.

That said, Reviewer 2 raises a valid deeper question: conditional on being retained, do viewers who are responsive to engagement graphs generate less advertising revenue than those who do not? We do not observe viewer-level behavior, so we cannot directly assess whether viewers who interact with the engagement graph differ in economic value from those who do not. We note, however, that the direction of any such difference is theoretically ambiguous. Viewers who use the engagement graph to navigate efficiently may generate less mid-roll ad exposure, as Reviewer 2 suggests; but they may also represent more intentional, higher-opportunity-cost viewers whom advertisers value, or they may engage more deeply with the content they do watch. Our data do not allow us to distinguish between these possibilities.

We acknowledge this in the revised paper’s discussion of economic impact (Section [X]). Our per-view revenue estimates assume that incremental views carry similar advertising value. If engagement-graph-responsive viewers are systematically less (or more) valuable per view, the economic estimates would be correspondingly adjusted. Resolving this would require viewer-level data on watch behavior and ad exposure, which we flag as a direction for future research.

3. Empirical strategy, identification, and inference

R2.3a. *It is worth investigating interference effects and potential SUTVA violations more. The main argument right now is that control videos are primarily from March and treatment videos are primarily from February and April so there should not be much interference, but this is very informal. Can you add more formal tests? One idea would be to compare the estimates in categories with a lot (less potential for interference) vs few (more potential for interference) videos, but the authors might have better ideas.*

Response: ...

R2.3b. *If most of the control videos are from March while treatment videos are from February and April, treatment assignment could be confounded with any differential trends affecting popular (50K+ views) videos in the end of March versus end of February or end of April. You can check whether this is an issue by balancing treatment and control samples of popular videos on which week these videos were uploaded (you cannot do days because an interaction of day * "popular video" defines treatment assignment).*

Response: We thank Reviewer 2 for raising this insightful point. We address this concern with two robustness checks. The adjacent upload date matching (response to R1.1a, Tables [X]) restricts all comparisons to videos uploaded within seven calendar days of each other, ensuring that treated and control videos within each matching stratum face similar macro conditions. The weekday/weekend matching robustness check (response to R1.1a, Tables [X]) addresses potential confounding from systematic upload-day patterns. Results remain consistent under both restrictions, indicating that the temporal composition of the treatment and control groups does not drive our findings.

R2.3c. *Inference: standard errors should be clustered on the level of the treatment assignment (Abadie et al 2023), which in this case are incidents of batch processing.*

Response: ...

4. Within-video effects

***R2.4a.** I have found the estimates in Section 6.1 (e.g. Figure 11) hard to interpret. Changes in the inequality in views on a video can be a result of the change of the composition of viewers over time, not a treatment effect of the engagement graph. In particular, consumers who find the video later on will be searching for or recommended this video because they are interested in the topic, not because YouTube recommended them this video for its novelty value.*

Response: Reviewer 2’s concern about compositional changes in the viewer pool over time is well-taken and mirrors the concern raised by the AE in Section 1.4. As noted in our responses to R1 (Mechanism) and AE 1.4 below, we have removed the within-video engagement concentration analysis (original Section 6.1) entirely from the revised paper. The compositional identification challenge is difficult to address cleanly with aggregate video-level data, and we chose to replace this descriptive analysis with the transcript-based heterogeneous treatment effects analysis, which operates at the video level using pre-determined content features.

***R2.4b.** More broadly, I find the effects of engagement graphs on the allocation of time that consumers spend within the video to be the most interesting in this context. First, it has important implications for the advertising exposure of consumers: if consumers skip ahead to the most interesting parts and watch less time overall, they are less likely to get exposed to midroll ads. Second, the ability of viewers to skip to the most interesting parts has implications for how video producers should design their products (e.g. Liu et al 2020).*

Response: We agree that the within-video allocation of viewer attention is one of the most important downstream question raised by our findings. We do not have viewer-level watch time data and cannot directly measure within-video attention allocation; we acknowledge this limitation transparently in the revised paper.

Our transcript-based heterogeneity analysis speaks to the content design implication Reviewer 2 highlights, albeit from a different angle. Rather than measuring attention allocation directly, we characterize which content structures are associated with the strongest engagement graph treatment effects. Our finding that content with deferred payoffs and lighter information delivery benefits most is directly relevant to how creators should think about content design in an environment where within-content engagement signals are visible. We are careful to note that these are *associations* between content characteristics and treatment effect magnitudes rather than causal claims about content design, but they provide empirically grounded directions that creators can evaluate in their own production decisions.

Associate Editor

Identification

AE 1.1. *Reviewer 2 raises an important concern regarding the identification and interpretation of the incremental effects of the engagement graph. Specifically, the R2.1c notes that users only become aware of the engagement graph after they have already started watching the video. Therefore, it might initially appear puzzling that the authors document an increase in both views and likes of similar magnitudes—if the engagement graph simply enhances viewer satisfaction or utility conditional on having already started watching, we would primarily expect increases in likes and comments, but not necessarily in the number of views.*

There seems to be a logical explanation consistent with the authors’ empirical findings, tied directly to the definition of a “view” on YouTube. As the authors clarify later in the paper (p. 29), a view is recorded when a user cumulatively watches at least 30 seconds of the video, including cases where viewers skip ahead but still accumulate at least 30

seconds of viewing time. (“Recall that a view is recorded when a user watches at least 30 seconds of a video cumulatively, including cases where users skip through the video but accumulate more than 30 seconds of total watch time. Hence, this result suggests that having the engagement graph successfully retains more viewers and increases video consumption, measured by views.”).

Thus, the presence of an engagement graph may encourage viewers who might otherwise abandon the video early to stay engaged longer—specifically, long enough to meet this threshold and be counted as an incremental view. This retention-based mechanism is likely what is driving the views result and should probably be explicitly highlighted earlier in the paper.

However, this also exposes a critical limitation of the current analysis. Due to data constraints, the authors can observe only aggregate video-level metrics (views, likes, comments) and do not have detailed viewer-level engagement data or viewing duration metrics. Consequently, the authors cannot directly assess how total viewing time changes in response to the engagement graph. This limitation is significant because, as R2 insightfully points out, the engagement graph might lead viewers to selectively skip to highlighted peaks, potentially decreasing overall viewing duration. Given these constraints, it is my understanding that the paper currently cannot empirically disentangle whether the engagement graph leads to an increase or decrease in total viewing time; we just know that it increases number of viewers who watch longer than 30 seconds.

I wonder if one potential angle to explore is whether variation in video lengths could somehow help address this limitation. Since a “view” is consistently defined as 30 seconds watched, this fixed threshold might introduce some exogenous variation in the fraction of total viewing time across videos of different lengths. For shorter videos, 30 seconds constitutes a larger share of total duration than for longer videos. It might be worth considering whether this variation could be leveraged in a creative way.

In my view, this is a critical limitation that directly influences the downstream im-

plications of the findings. The authors should explicitly acknowledge that while their analysis credibly demonstrates incremental views (crossing YouTube’s 30-second threshold), the absence of data on total viewing duration prevents them from determining whether overall viewer engagement with a video increases or decreases. This limitation is particularly important because it directly affects implications regarding ad exposure and monetization for content creators and the platform and introduces significant uncertainty about the contribution of this paper.

Response: We are grateful to the AE for articulating the retention-based interpretation clearly. We adopt this framing in the revised paper and highlight it earlier in the paper.

We also transparently acknowledge the data limitation the AE identifies: because we observe only aggregate video-level metrics, we cannot directly measure total viewing duration or disentangle whether the engagement graph increases or decreases overall watch time. This is stated clearly in the revised paper’s limitations discussion.

On the AE’s suggestion about video length variation: we include video length as a continuous moderator in our heterogeneous treatment effects specification. Longer videos show somewhat larger engagement graph treatment effects on views and likes, consistent with the intuition that longer content presents greater viewer uncertainty about whether the video is worth watching in full, and therefore benefits more from an engagement graph that signals value. We discuss this finding in Section [X].

AE 1.2. *R2.1b raises an important concern regarding the potential role of YouTube’s recommendation algorithm. Even small initial changes in viewer engagement triggered by activation of the engagement graph could subsequently prompt algorithmic adjustments, increasing the video’s visibility. Such algorithmic responses complicate disentangling direct viewer reactions from indirect, algorithmically-driven effects.*

An important conceptual issue here is how exactly the treatment effect is defined. If the engagement graph directly increases engagement, which in turn triggers algorithmic

mic boosts, the subsequent increases in views could be considered indirectly downstream causal—these additional views would not have occurred without the initial engagement graph effect. Given these complexities, the authors should transparently acknowledge this issue, clearly stating that their causal estimates likely capture both direct effects (viewer responses) and indirect effects (algorithmically mediated). Note that this issue also creates limitations for interpretation, as the estimated effects depend on the specific recommendation algorithm in place at a given time. Small algorithmic adjustments could substantially alter the observed outcomes, limiting the generalizability of these findings. Overall, I encourage the authors to be very transparent about the limitations in generalizability of the results.

Response: As discussed in our response to R2.1b, we agree that our treatment effect plausibly captures both direct viewer responses to the engagement graph and indirect effects mediated by YouTube’s recommendation algorithm, and we explicitly acknowledge this in the revised paper. Importantly, we argue that the combined effect, i.e., direct viewer response plus any algorithmically mediated downstream effect, is the managerially relevant estimand. From the perspective of platforms and content creators, what matters is the total effect of the engagement graph’s activation on video performance, taking as given the current market conditions and algorithms. We frame the combined effect as such in the revised paper. We acknowledge that the exact magnitude of the estimated effects may be specific to YouTube’s recommendation algorithm at the time of our study, and that different algorithmic regimes could yield different magnitudes of indirect effects. We discuss this limitation in generalizability explicitly in the revised discussion section.

AE 1.3. *Reviewer 1 correctly notes that the identification strategy, while compelling, depends critically on the assumption that the timing of video uploads is uncorrelated with unobserved video quality or underlying market conditions (R1.1, R2.3). Although the authors partially address this through matching and the difference-in-differences frame-*

work, R1’s concerns could be more convincingly addressed with further analysis of parallel trends and additional placebo tests. For example, the current event-study analyses are conducted at a weekly level with only two pre-treatment periods, despite the main analysis using 48-hour periods. Given the known state dependence in viewing behavior (likely interacting with YouTube’s recommendation algorithm) demonstrating parallel trends at the same granularity as the main analysis (i.e. 48-hours) would substantially strengthen the identification (this might mean that the pool of control videos might need to change). This would increase the number of pre-treatment periods from two to approximately seven and provide stronger evidence of parallel evolution in outcomes. Additionally, the authors could consider adding another placebo test that artificially assigns treatment entirely within the pre-treatment period.

Response: We address the identification concerns raised by R1, R2, and the AE through multiple analyses. Regarding the concerns about upload timing and temporal confounding (R1.1, R2.3b), we conduct robustness checks using weekday/weekend matching (Tables [X]), adjacent upload date matching (Tables [X]), and unmatched sample estimation (Tables [X]), as described in our responses to R1.1a and R2.3b above. Here we focus on the AE’s two specific suggestions regarding parallel trends and placebo tests.

We first wish to clarify that our event study periods are already 48-hour intervals (not weekly), consistent with the main analysis. The constraint limiting us to two pre-treatment coefficients in the baseline stems from both our matching design and the institutional features of the setting. On the matching side, we match on video initial growth trend using both $p = 0$ and $p = 1$ data, which excludes these periods from estimation, and one additional period must serve as the event-study baseline. On the institutional side, although we track each video for more than three weeks, most videos reach their eligibility date within the first week after upload—meaning treatment activates early in the video lifecycle. The number of available pre-treatment periods is therefore structurally bounded by when YouTube’s batch processing assigns eligibility, which is outside our control.

Extended pre-treatment window. With this constraint in mind, we extend the pre-treatment window as far as the setting allows. We implement an alternative matching design that matches on $p = 0$ data only (cumulative views, cumulative likes, video duration, and content category), freeing up $p = 1$ as an additional pre-treatment estimation period. To partially compensate for the loss of initial trend information in the matching step, we add an exact-match requirement on upload weekday/weekend, which captures systematic differences in early engagement dynamics associated with upload timing. This yields a sample of 5,243 videos with four pre-treatment estimation periods and three pre-treatment coefficients (one period serves as the baseline) at the 48-hour granularity of our main analysis.

The pre-treatment coefficients across all three outcomes are overall statistically indistinguishable from zero (Figure [X]), providing strengthened evidence for the parallel trends assumption. Post-treatment effects remain significant and comparable to baseline (CS: 7.2% on views, 6.4% on likes, 2.75 additional comments, all $p < 0.001$). Results are reported in Tables [X] and [X] and Figure [X] in the revised paper.

Placebo test with artificially assigned treatment timing. Following AE’s suggestion, we conduct a complementary placebo test that is independent of the control group. We use only the originally treated videos from our matched sample and randomly split them into “fake-treated” and “fake-control” groups, then artificially assign treatment timing within the pre-treatment window. Under this design, any estimated effect of the placebo treatment would indicate a failure of the parallel trends assumption rather than a true treatment effect. Results show that all placebo estimates are small and statistically indistinguishable from zero across both CS and TWFE estimators and all three outcomes (Tables [X] and [X]), confirming that the main findings are not driven by spurious differential trends in the pre-treatment period.

AE 1.4. *R2.4a also highlights a concern that complicates the causal interpretation of the within-video analysis. Specifically, the observed shift in viewer attention toward*

highlighted peaks after the engagement graph activation might reflect changes in the composition of viewers rather than a direct causal effect on individual viewer behavior. If the engagement graph disproportionately attracts viewers who prefer skipping or selectively watching specific video segments, then the observed within-video concentration effects could arise from viewer selection rather than observational learning per se. This potential compositional shift makes it difficult to conclusively attribute the within-video findings purely to the causal effect of the engagement graph feature. To the extent possible the authors should try to address this concern. If a clean identification is not feasible, the authors should transparently acknowledge this limitation and frame these findings as providing valuable descriptive insights rather than definitive causal evidence. Also note that R1.2 has many thoughtful comments about observational learning as a mechanism that should be addressed.

Response: As discussed in our responses to R1 (Mechanism) and R2.4a, we have removed the within-video engagement concentration analysis and the observational learning mechanism section entirely from the revised paper. We replace this analysis with the transcript-based heterogeneous treatment effects analysis. This new analysis operates at the video level using pre-determined content features and is not subject to the compositional confounds that affected the original within-video descriptive results.

Contribution and positioning

AE, Contribution: *R2.4b convincingly argues that the authors should substantially refine their discussion of managerial implications. Currently, the paper understates complexities arising from the fact that video producers have limited direct control over when and whether the engagement graph is activated, creating significant uncertainty regarding the contribution of the paper. However, I am optimistic that the authors can reposition the paper with additional analyses that could strengthen the contribution. Below, I sug-*

gest potential avenues for strengthening the managerial relevance of the paper—though I acknowledge that the authors are in a better position to potentially develop even stronger ideas. Broadly, the revised manuscript would benefit from explicitly considering implications in the following areas (or a subset of them): (1) ad avoidance, (2) economic outcomes for creators, and (3) optimal video content design.

Specifically, the authors could potentially:

- *Leverage the observed distribution of engagement peaks to run simulations evaluating hypothetical yet realistic scenarios for mid-roll ad placements.*
- *Explicitly acknowledge potential strategic responses by YouTube, such as dynamically placing mid-roll ads immediately before high-engagement segments, which could partially offset viewer skipping behavior.*
- *Another promising—but admittedly challenging—direction (relating to points 2 and 3) would be to systematically characterize the type of content that drives engagement peaks. If feasible, the authors could consider applying computer vision or related techniques to classify peak segments according to features such as visual complexity, emotional intensity, narrative structure, or pacing. While this approach is clearly demanding, these insights into systematic patterns across categories would significantly strengthen the paper’s contribution. Relevant references here include Simonov et al. (2023) on suspense and surprise in media product design, as well as Liu et al. (2020) (R2.4b).*

Response: We have substantially restructured the paper’s contribution in response to this feedback and the broader editorial dialogue. As noted in the introduction to this response letter, the revision followed three rounds of proposals and feedback with the editors. The AE’s final assessment of our revision proposal confirmed the following scope:

“In my view, the paper can be framed cleanly as follows: (1) a demand-side main effect documenting that engagement graphs change viewer behavior; (2) demand-

side heterogeneity showing that the effect is systematically larger or smaller for certain types of content, which the authors can characterize using transcripts that they already have; and (3) a supply-side analysis examining whether creators respond to the rollout along margins that do not require transcripts.”

We have executed the revision along these three pillars. We note that we ultimately acquired supply-side transcripts and conducted content-structure analyses on the supply side as well, going beyond the minimum scope of the revision contract, because we judged these analyses would meaningfully strengthen the paper. Below we describe each pillar in turn, focusing on the key design choices and headline findings; full details are in the revised paper.

(1) Demand-side main effect with strengthened identification. We document the causal effect of engagement graphs on viewer behavior (views, likes, comments) with substantially strengthened identification, as described throughout this response letter. The set of robustness checks, including weekday/weekend matching, adjacent upload date matching, unmatched sample, extended pre-trends, and placebo tests, collectively address the identification concerns raised by both reviewers and the AE.

(2) Demand-side heterogeneity via transcript-based content features. In place of the original mechanism analysis, we extract content features from video transcripts and estimate how these features moderate the treatment effect of engagement graphs. This analysis directly addresses the AE’s call to systematically characterize which types of content are most affected by the engagement graph, as well as the content design direction suggested by the AE and R2.4b.

We extract segment-level content features from video transcripts using an LLM, classifying each one-minute segment along an *information dimension* (payoff/reveal presence, information density, new concept introduction) motivated by cognitive load theory (Sweller, 1988; Mayer, 2002) and a *narrative dimension* (structural role) grounded in classical narrative theory (Freytag, 1895). This yields 26,021 segments across 2,186 treated videos. We validate these classifications descriptively: feature distributions across video position follow

recognizable structural patterns (e.g., introduction segments dominate the opening, climax segments peak in the 65–85% range), and substantive content features are positively associated with segment-level viewer engagement (e.g., payoff segments show meaningfully higher mean engagement than non-payoff segments). These associations are purely descriptive rather than causal, but they provide face validity for the LLM classifications.

We aggregate to the video level along four dimensions: *payoff lateness* (where in the video value delivery occurs), *information density* (how substantively packed the content is), *percentage of new concepts* (how often the video introduces novel material), and *narrative completeness* (the extent to which the video follows a conventional dramatic arc). The first three are informational characteristics; the fourth is structural. We include all four as continuous moderators, along with video length and channel popularity, in our heterogeneous treatment effects specification, with all moderators mean-centered and standardized.

Our conceptual framework centers on a conjecture about *value uncertainty*: the engagement graph should be most valuable when a prospective viewer faces greater difficulty assessing whether the remaining content is worth watching. Along the informational dimension, we hypothesize that the engagement graph matters more when payoffs arrive late (the viewer has not yet seen the value they came for), when content is informationally lighter (the content is less self-revealing about its own worth), and when new concepts are introduced frequently throughout the timeline (it’s harder for the viewer to predict what future segments will cover). Along the structural dimension, we include narrative completeness to test whether formal content organization independently moderates the effect beyond informational characteristics.

The results support the informational-value interpretation. All three informational features significantly moderate the treatment effect on views and likes in the predicted directions: payoff lateness is a positive moderator, information density is a negative moderator, and percentage of new concept is a positive moderator. Narrative completeness, by contrast, does not significantly moderate the effect on any outcome.

We highlight two advantages of this content-feature analysis relative to the original peak-count heterogeneity. First, the content dimensions we identify are features that creators actively decide about when producing videos, giving the heterogeneity results practical relevance for content design that graph-shape characteristics lack. We are careful to note that these are *associations* between content characteristics and treatment effect magnitudes rather than causal claims about content design, but they provide empirically grounded directions that creators can evaluate in their own production decisions. Second, these findings are more likely to generalize across platforms that surface within-content engagement signals, since they characterize the underlying content structure rather than a platform-specific feature implementation.

(3) Supply-side analysis. We conduct a new difference-in-differences analysis exploiting the May 2022 platform-wide rollout of engagement graphs to examine whether and how creators respond to the changed information environment. We leverage the retroactive activation of engagement graphs on all eligible pre-rollout videos, which created differential exposure across creators based on the proportion of their pre-rollout videos that qualified for the feature. We compare high- versus low-exposure creators (median split) before and after the rollout in a sample of 3,864 channels spanning 13 content categories. We examine both content supply decisions (upload frequency, video length) and content structure (information density, payoff location, narrative role proportions), the latter using transcripts collected for over 300,000 videos in the supply-side sample.

The main finding is that high-exposure creators do not substantially alter their upload volume, video length, or content structure over the four-month post-rollout window we examine. All estimates are small and statistically indistinguishable from zero, with one marginal exception: a small decline in conclusion segment share among high-exposure creators ($\hat{\beta} = -0.0018$, $p = 0.054$), which we interpret cautiously given its marginal significance and the multiple outcomes tested.

This largely null supply-side result is itself a substantive finding. It indicates that creator

production behavior is relatively insensitive to incremental changes in the platform information environment over the horizon we examine, suggesting that direct incentive mechanisms—such as revenue sharing arrangements—may be more effective levers for influencing creator behavior than changes to the viewer-facing information environment alone. Full details of the supply-side research design, data construction, and results are in Section [X] of the revised paper.

Additional comments

***AE, Additional:** Both reviewers provided several other useful comments that could further strengthen the paper. In particular, R2.3 raises important methodological concerns about potential violations of SUTVA and clustering of standard errors that should be carefully addressed in the revision.*

Response: Both the SUTVA and clustering concerns are addressed in our responses to R2.3a, R2.3b, and R2.3c above.

We are grateful to both reviewers and the editorial team for the thoroughness and care of the review process. We believe the revised paper makes a substantially stronger contribution on all dimensions: identification, heterogeneity, and supply-side analysis. We look forward to the editorial team’s response and welcome any further questions.

Sincerely,

The Authors

References

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Appendix